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RESEARCH ARTICLE

Phase-Sensitive Dynamic Filters: A Conceptual Model for Energy-Preserving, Instantaneously Tunable, Symbol-Level Signal Processing Using Complex IIR Filters

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ABSTRACT Filters used for modulated RF signals are often matched bandpass filters that have larger bandwidths than that of the modulation. Therefore, the time constants of the filters must be shorter than the symbol periods. This paper presents an idealized theoretical model for Phase-Sensitive Dynamic Filters (PSDF), a class of emerging filters that can be tuned in real time, as fast as the symbol rate of a signal but with time constants longer than the symbol periods. With the foreknowledge of the frequency and phase shifts of a given signal, such filters are capable of filtering across multiple symbols of a constant envelope and compressing the noise bandwidth to a fraction of the modulated signal bandwidth. The transient response of the filters can be longer than the symbol rate of the signal. This can mathematically be modeled by infinite impulse response (IIR) filters with long time constants yet fast time-varying complex coefficients in the feedback path. Phase-sensitive dynamic filters preserve energy across symbols and also exhibit phase-sensitivity. The proposed models indicate that PSDFs can be employed in applications such as rejection of self-interference in the front ends of full-duplex communications systems. Such systems pave the way for the possibility of higher isolation and robustness compared to other prototypical approaches in dynamically changing systems.

INDEX TERMS Charge-preservation, complex filters, dynamic filters, IIR filters, phase-sensitive, time-varying, tunable filters.

I. INTRODUCTION

Fixed bandpass filters play an integral role in virtually all communications systems. However, their fixed nature often clashes with the increasing need for dynamically changing frequency and bandwidth resources. With this problem arises the need to be able to tune filters in real time. Tunable filters serve the primary advantage of being able to change the covered frequency bands without the need to physically reconfigure the hardware. The advantages of such filters

cannot be understated; aligning to the desired frequency in frequency hopping, noise-minimization and spectral suppression of undesired and unpredictable tones and interference, and software-defined or cognitive radios are examples in which the necessity of tunable filters is demonstrated [1], [2], [3], [4]. However, oftentimes the conventional tunable bandpass filters are inadequate for real time tuning due to their slow switching speeds and inability to track phase and frequency changes in real time, limiting their usefulness in systems requiring faster transition times.

It is well-known that the three fundamental characteristics which make up a signal – frequency, phase, and

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amplitude – can be used towards modulating information into the signal. For this paper, the focus will be on the former two characteristics – frequency and phase. Assuming a signal has both varying phase and frequency (e.g. frequency-hopped QPSK, phase-discontinuous BFSK), the traditional fixed-filter approach is to design a bandpass filter which encapsulates the entirety of the spectrum in which the signal is located. This approach is straightforward and easy to implement in both the digital and analog domains. However, with this approach, excess noise is captured outside the bandwidths of the region of signal power. For the simple case of a BFSK signal which switches its transmitted carrier frequency across two fixed values, the excess noise can be visualized in Fig. 1(a). To tackle this, there have been many approaches to develop filters that can switch between specific passbands in both the analog and digital domain, such as [2], [5], [6], [7]. These methods do an adequate job of filtering out the excess noise that would otherwise be captured by a traditional fixed-bandpass approach, as visualized in Fig. 1(b).

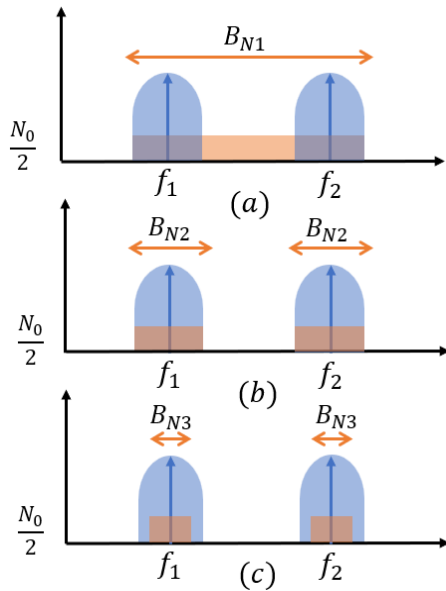


FIGURE 1. Comparison of filter bandwidths across (a) fixed filters, (b) switched filter banks, (c) and phase-sensitive dynamic filters. Excess noise characterized by highlighted orange within each spectrum for each filtering approach.

One such method, so-called “*time-frequency peak filtering*” [8] provides a correlation-based method to accomplish this form of noise-suppression. Another popular approach is the switched filter bank method [9], [10], [11], in which multiple passbands filters are all constructed and switched to, based on the desired frequency. For transmitted signals in which the carrier frequency is hopped between predetermined values, these approaches cannot switch at rates faster than the symbol rate of a signal due to the direct inverse relation between the settling time of a filter and its bandwidth. The relation between bandwidth and a filter’s time constant can

be given by the following equation [12]:

$$\Delta f \cdot \tau_f = \frac{1}{2\pi} \quad (1)$$

where Δf is the 3-dB passband of the filter, and τ_f is the time constant (often chosen to be modeled by RC in circuit theory). The Q factor of the filter is given by the following equation:

$$Q = \frac{f_c}{\Delta f} \quad (2)$$

where f_c is the center frequency of the filter. From this, it can be observed that as the bandwidth of a filter increases, the time constant decreases, and the Q factor decreases – thus indicating a limit in time it takes a filter to settle given its operating bandwidth. The higher the Q , the longer it takes for the signal to settle. Depending on the application, higher- Q filters, or filters with narrower bandwidths, are desired for signals in close proximity together. However, with this comes the cost of elongating the settling time. From a communications perspective, the filter is only deemed operational when it has fully “settled” (time constant reached). For fixed bandpass filtering, this does not present a critical concern, as it is merely a fixed latency which takes place in the initial portion of filtering, which does not have any significant drawback. However, for applications seeking real time symbol-level switching, this is a cause of concern as this bandwidth limit introduces periods of inactivity due to the intrinsic settling time between switching, effectively distorting the signal and decreasing the spectral efficiency if the filter response time is comparable to or longer than the symbol periods.

The critical role of high- Q resonators has been highlighted through recent progress in parametric mixers and tunable RF-photonics filters. Instantaneous mixing behavior and dynamic tunability has also been exploited by monolithic RF implementations, such as directly pumped time-varying transmission lines (TVTLs) and integrated parametric mixers [13], [14], [15], [16], [17]. Conversely, photonic systems – like dual-comb RF filters and cascaded interferometric leverage high- Q optical cavities to enable rapid passband reconfiguration filters [2], [18], [19], [20]. Both areas of research showcase charge conservation and phase coherence to achieve high-efficiency frequency conversion and dynamic passband control. While prior work has advanced these domains independently, a key limitation still persists: modeling phase-sensitivity and energy preservation.

In this work, we propose a unified physical model of phase sensitive dynamic filters (PSDFs), as well as some of the traditional concerns regarding spectrum utilization and rapid frequency/phase shifts. They can be directly implemented in the RF domain with filters that utilize charge accumulations [21], [22], [23], [24] or energy accumulations such as the previously cited parametric work. They can be tuned in real time, with switching speeds faster than the symbol rate of a signal while maintaining time constants longer than the symbol periods. The property of energy/charge preservation across symbols allows dynamic filters to surpass the fundamental limit of the transient duration and bandwidth. With the

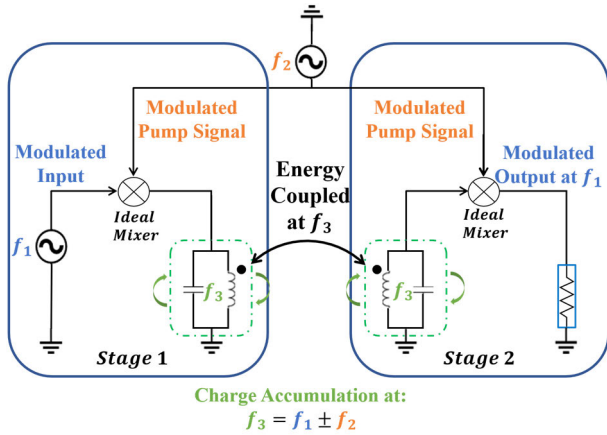


FIGURE 2. Ideal circuit model of a dynamic filter in its simplest form.

foreknowledge of the signal phase/frequency shifts readily available, dynamic filtering can filter signals across varying frequencies/phases and even further minimize the noise to a bandwidth corresponding to that defined by the time constant of the filter without the constraint of “resettling” within a symbol period. The effect this has on noise-suppression can be seen in Fig. 1(c). Such filtering systems, if physically realized, offer a great deal of applicability and robustness in the front end of RF systems on top of the known advantages offered by traditional tunable bandpass filters.

1) First, phase-sensitive dynamic filters offer phase-sensitivity and selectivity. When designed to have a finite bandwidth, the coefficients can be configured and switched in such a way that only a particular phase of a given frequency is captured.

2) Second, dynamic allocation of transmitting and receiving bands is possible with these filters. This advantage is particularly applicable to frequency-hopped or adaptive radio systems and is especially useful in applications with fluctuating frequencies of operation, without compromising spectral efficiency due to the frequency-bandwidth limit.

3) Third, PSDFs can be operated in a way that offers increased robustness against external noise due to the “correlative” nature of charge preservation.

4) Finally, with knowledge of the transmitted waveform, dynamic filtering offers selective transmitter leakage suppression without compromising received information in transceiver systems. For PSDFs implemented in RF circuitry, the implication of this is the removal of complicated RF leakage cancellation circuitry and in-band full-duplex communications. This will be further discussed in this paper.

This work builds upon the author’s earlier studies on dynamic filtering [17], [25], [26]. Here, a generalized theoretical framework is developed to model the broader class of charge and energy-preserving RF signal-processing technologies, while incorporating notch-filter behavior, spectral characteristics, and noise analysis within a unified idealized system model. The analysis in this work adopts a

system-level abstraction that intentionally neglects circuit-level non-idealities such as switching transients, timing jitter, and pump phase noise in order to highlight the potential utility of such filters and to clarify the fundamental operating principles that enable phase-sensitive dynamic filtering. Earlier work demonstrates experimentally realizable parametric filtering architectures based on frequency translation, in which charge and energy are redistributed through controlled interactions with pump frequencies, and serve as concrete motivation for the idealized model considered here.

To fully explore the potential of PSDFs in communication systems, it is imperative to develop an intuitive mathematical model. In this paper, we propose complex IIR filters to serve as an adequate model in demonstrating the theory behind the phase-sensitive dynamic filters because of their recursive nature. This mirrors physical charge-preserving systems while addressing phase-sensitivity gaps in prior work. The paper first reviews the traditional recursive IIR filter theory and extends it to include symbol-synchronous dynamic switching of complex coefficients. A theoretical framework is then used to demonstrate the behavior of the equivalent PSDF across two different types of commonly used signal schemes (BFSK and QPSK), highlighting both frequency and phase sensitivity. Finally, a system-level simulation of a front-end leakage-suppression scheme is introduced to illustrate the potential of such a device.

II. THEORY OF PHASE-SENSITIVE DYNAMIC FILTERING

The ideal circuit model in Fig. 2 builds upon a modified mixer framework and serves as the analog counterpart of the proposed PSDF. This model demonstrates instantaneous tunability through active pump modulation, which can be observed by varying the pump’s frequency and phase in real-time. In practice, the loss penalty of the mixers can be removed with the recent developments of mixer first approaches [22], [27], [28], [29]. Although the analysis in this section is carried out purely at the theoretical level, the idealized time-varying complex IIR model considered here is motivated by frequency-translation-based parametric filtering architectures that have been experimentally demonstrated, such as the cascaded parametric mixer realization reported in [17]. In this case, the recursive nature of IIR filtering inherently mimics the charge accumulation and energy preservation showcased in the circuit.

The procedure is as follows: a phase/frequency modulated input signal at frequency f_1 is mixed with the pump signal at f_2 that contains identical phase/frequency transitions. The resulting mixed signal at f_3 forms a resonance corresponding to a single sinusoid. This accumulated energy is then coupled into the subsequent stage, where the same signal at f_2 modulates the sinusoid to produce the final output back to the original frequency at f_1 . This process effectively decouples the achievable bandwidth from traditional RC time constant limitations while incorporating phase-sensitive feedback to correct passband distortion while conserving charge.

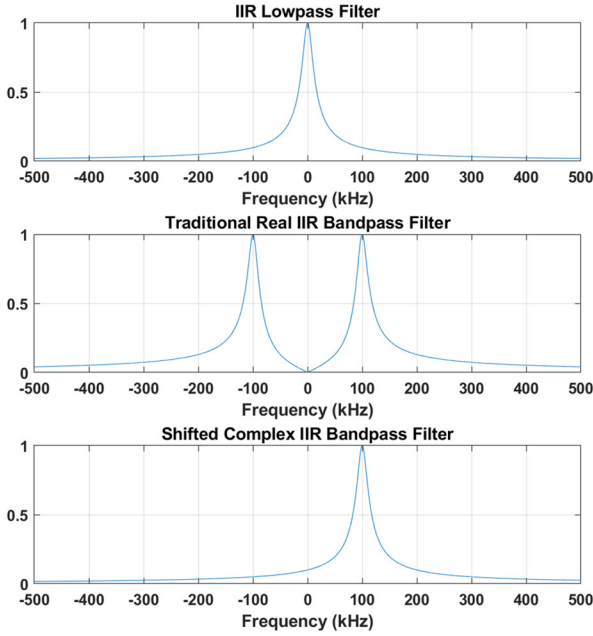


FIGURE 3. Comparison of frequency responses across lowpass filters, traditional bandpass filters, and complex IIR filters.

To formalize this behavior mathematically, the system can be abstracted as a time-varying complex IIR filter. While IIR filter coefficients are typically restricted to be real values to produce real filter systems, complex coefficients may also be employed and offer distinct advantages. The foundation for such filters was introduced by Crystal and Ehrman [30], who explored the “shifted” nature of complex filters and demonstrated that complex coefficients can directly realize bandpass responses without requiring conventional lowpass-to-bandpass transformations. Subsequent research has examined computational complexity, frequency-response characteristics, and overall performance of complex filters. However, existing complex filtering systems are largely fixed (except for some adaptive methods [31], [32], [33]). To the authors’ knowledge, there has been little prior exploration of dynamically reconfigured complex-coefficient systems in which the coefficients are switched in real time.

A. FILTER MODEL WITH DYNAMIC FREQUENCY TUNABILITY

Recursive filters are filters that are implemented in the time domain. Compared to traditional convolutional filtering, recursive filters provide an effective way of achieving a long impulse response without having to perform a long convolution. This is essential for PSDFs, as long time constants are desirable. IIR filters usually take the following form:

$$y[n] = \sum_{i=0}^A a_n x[n-i] + \sum_{i=1}^B b_n y[n-i]. \quad (3)$$

In contrast to IIR filters, finite impulse response (FIR) filters do not contain the secondary feedback term and thus lack an

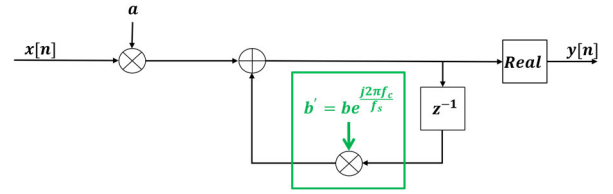


FIGURE 4. Representation of first-order complex IIR filter with frequency-tunability in the feedback term. Adapted from [25].

infinite response. As such, their filtering time is generally short and will not be considered in modeling this class of filter. For the purposes of this paper, only a single-pole complex IIR filter is considered. While the framework naturally extends to higher-order implementations through cascaded or multi-pole structures, the single-pole case is presented here to simplify the exposition and provide clear physical and mathematical intuition.

Typically, single-pole IIR filters are utilized for low-pass or high-pass filtering, but we observe that the usage of complex variables enables us to shift the pass-band frequency of a low-pass filter up to a specific passband frequency. The typical single-pole digital IIR filter takes the following form:

$$y[n] = ax[n] + by[n-1]. \quad (4)$$

For single-pole IIR filters, the values of a and b are chosen based on the desired normalized cutoff frequency and are traditionally held to observe the following relationship [34]:

$$b = e^{-2\pi f_{cutoff}} \quad (5)$$

$$a = 1 - b \quad (6)$$

where f_{cutoff} is the normalized cutoff frequency of the low-pass filter. Although (5) is not a necessary condition for IIR filtering to take place, it serves as a fundamental condition in which other characteristics (such as the cutoff frequency) can be derived. Using the z-transform of the one-pole filter, the transfer function can be solved algebraically:

$$Y[z] = aX[z] + bY[z]z^{-1} \quad (7)$$

$$H[z] = \frac{Y[z]}{X[z]} = \frac{a}{1 - bz^{-1}}. \quad (8)$$

By substituting in $z_{shift} = e^{-j\omega_c T} z$ into (8), we can effectively shift the passband of the frequency from 0 to ω_c . This leads to:

$$H_{complex}[z] = \frac{a}{1 - be^{j\omega_c T} z^{-1}} = \frac{a}{1 - b'z^{-1}} \quad (9)$$

$$b' = be^{j\omega_c T} = be^{j2\pi \left(\frac{f_c}{f_s}\right)} \quad (10)$$

where b' is a complex coefficient and $T = 1/f_s$ is the sampling interval. Using the inverse z-transform, the impulse response of the complex IIR filter can also be solved to be:

$$h[n] = a \cdot b'^n u[n]. \quad (11)$$

Such filters are asymmetric about the zero-frequency axis (something characteristic of complex filters) in the frequency domain. The center point of the filter is shifted according to ω_c . When starting with a real low-pass IIR filter, the symmetry and bandwidth of the entire frequency response about the center frequency is preserved, something not observed in traditional lowpass to bandpass transformations. The final response resembles the positive frequency axis of a traditional bandpass filter that is transformed from the corresponding low-pass prototype. Because of this, the original cutoff frequency of the lowpass filter becomes half the 3dB bandwidth (edge-to-edge) of the shifted complex bandpass filter - this can be seen in Fig. 3. However, in dynamic filtering specifically, only the time domain recursive filtering is utilized because it can be adjusted in real-time. Finally, as demonstrated in [35], only the real portion of the output is considered.

For this example, since the real equivalent of $H_{complex}$ is characterized by the sum of $H_{complex}$ and its complex conjugate, $H_{complex}^*$, the real portion of the output from $H_{complex}$ matches the desired output, $y[n]$, from that of the real equivalent. The imaginary component can be neglected for the final output, indicating that only a single complex filter is adequate to compute the output of the equivalent real filter. Additionally, we can also send a complex signal consisting of only the positive frequencies of its real equivalent, and the same process will produce the same output. The first order representation of such a filter can be seen in Fig. 4.

B. FILTER MODEL WITH DYNAMIC PHASE SHIFTING

Phase-modulated signals can be found in virtually all currently operational wireless systems. However, they run into issues with narrow passbands – namely, distortion [36]. This is typically a concern when the bandwidth of the transmitted signal is wider than the passband of the filter, even with the same center passband frequency. Transmitting a signal through such a channel results in the removal of key frequency components in the transitions between phases. Generally, in order to preserve the sharp phase shifts of a phase-shift-keyed (PSK) signal, the passband of the filter must be increased to sizes larger than desired bandwidth in order to capture those transitions. Because of this, more of the spectra is admitted into the receiving system, resulting in more noise, and undesired frequencies. Pulse-shaping could be used to remedy this; however, the transitions between phase shifts are still negatively impacted.

In the same way that the center pass-band frequency for complex IIR filters can be switched by utilizing the feedback term, we will show that the phase of the filter can also be switched to match that of the desired input signal using the feedforward term. The first part involves “nullifying” the phase of the input signal through the feedforward term. This is more clearly visualized in the time domain by

$$y[n] = a'x[n] + by[n-1] \quad (12)$$

$$a' = e^{-j\phi}a \quad (13)$$

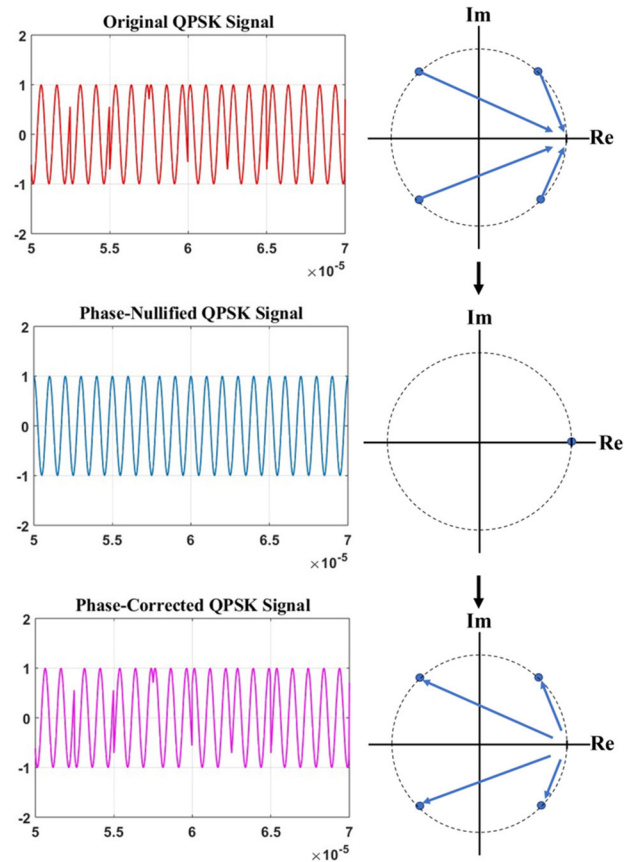


FIGURE 5. Effects of nullifying phase for a QPSK constellation. Two-step process involving phase-nullification at the input and phase-correction at the output. Adapted from [25].

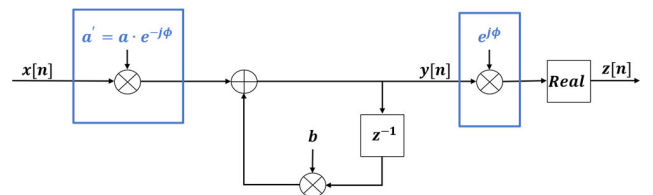


FIGURE 6. Representation of phase-sensitive portion of complex filtering. Adapted from [25].

where ϕ is the temporal instance of the phase in the incoming signal, and a' is the complex coefficient corresponding to the desired phase shift. It can be observed from (12) that, in contrast to complex frequency filtering, the complex coefficient is in the feed-forward term rather than the feedback term. The constellation of a traditional QPSK signal is plotted and can be seen in Fig. 5 above. The arrows indicate the rotation effects that the phase shift, ϕ , have on the constellation of the filtered waveform – mapping completely to the positive real domain. As can be observed in the same figure, the time-domain of the phase-corrected QPSK signal is compressed to simply a sinusoid, as all the different phases have been matched to one shared phase, thus reducing the bandwidth of the signal while preserving the bandwidth of the

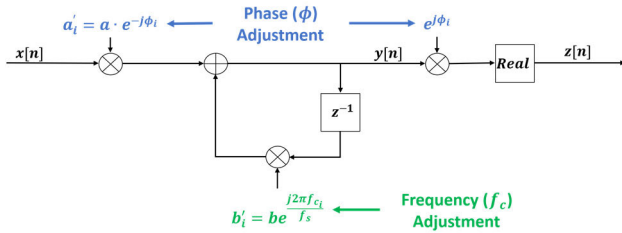


FIGURE 7. Full phase-sensitive dynamic filtering functional block diagram. Adapted from [25].

passband. The final portion of this filtering process involves re-correcting the phase. This can be simply done by adding the same exponential multiplier (with inverted phase) to the output of this filtering process. Once more, only the real component is considered. The relation is as follows:

$$z[n] = \text{Re} \left\{ e^{j\phi} y[n] \right\}. \quad (14)$$

The entire phase-sensitive filtering process can be seen in Fig. 6.

C. FULL PHASE-SENSITIVE DYNAMIC FILTER MODEL

The full dynamic filtering procedure can be visualized in Fig. 7 which is to apply the aforementioned frequency and phase shifting processes dynamically. In digital representations, the recursive filters become “dynamic” through the changing of the phase of the feedback coefficients which dictate the frequency of the filter passband and the phase of the feedforward coefficients that enacts a “nullification” and “reconstruction” effect on the overall phase of the signal. The amplitudes of the coefficients determine the bandwidth of the overall response. Implementing filters such as these are straightforward in the digital domain. One must swap the coefficients of the filter in accordance with the desired temporal switching requirements, and this is done in real time.

The number of sets of coefficients depends on the number of unique passbands and/or phase shifts that are desired:

$$y[n] = \sum_{i=0}^A a'_n x[n-i] + \sum_{i=1}^B b'_n y[n-i] \quad (15)$$

$$n \in \{1, 2, \dots\}$$

where a'_n and b'_n are the complex coefficients. Regarding the current single-pole complex IIR filter being explored, the recurrence equation will take the following form:

$$y[n] = a'_i x[n] + b'_i y[n-1]$$

$$a'_i = a e^{-j\phi_i}, b'_i = b e^{j2\pi f_{ci} T_s}$$

$$z[n] = \text{Re} \left\{ e^{j\phi_i} y[n] \right\}. \quad (16)$$

Evidently, as these complex coefficients are changed, the passband is also changed corresponding to the pair of chosen coefficients. The key here for dynamic filtering is that

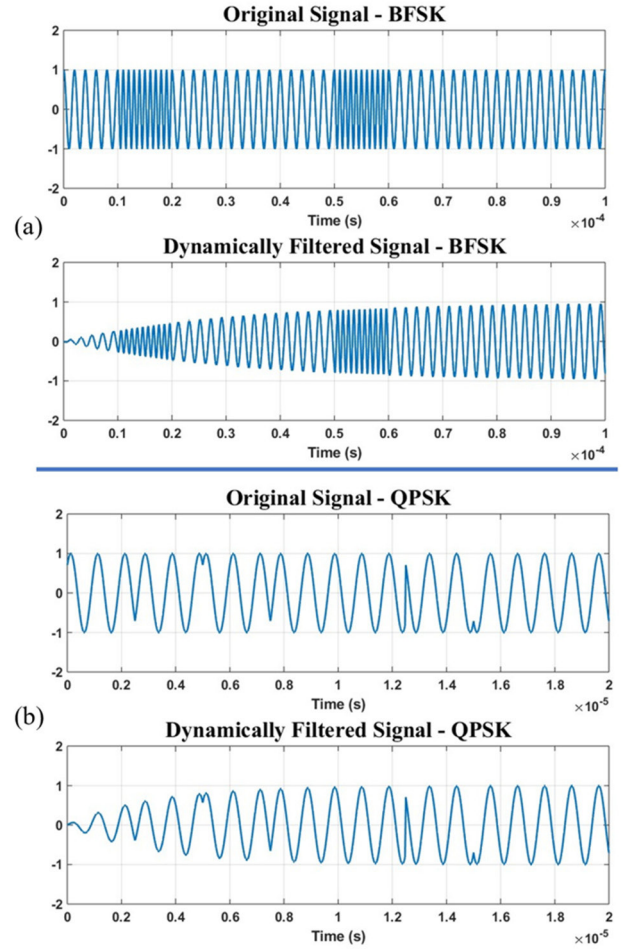


FIGURE 8. (a) Continuous charging behavior of modulated BFSK signals and (b) QPSK signals in dynamic filtering.

the “charge” is conserved across frequency shifts. In other words, the filter does not need to “resettle” to a new passband, which is something not typically found in traditional tunable bandpass filters. This is because settling time (or charging time) is dependent on the time constant of the filter, which is then dependent on the magnitude of the coefficients, not the phase. This implies that merely changing the phase of the coefficients not only leaves the time constant unchanged, but also removes the “resettling” effect imposed by the time constant due to changing input phases. Or more mathematically put, for IIR filters it follows that [34]:

$$|b| = e^{-\frac{1}{d}} \quad (17)$$

where d is the time constant of the single-pole complex filter. This is important for real time switching configurations, as the settling time is directly related to the selected bandwidth of the complex IIR filter. The time constant must be much greater than the symbol period to preserve the energy between shifts, which is stored in both the real and complex parts of the signal. This continuous charging effect for QPSK and BFSK signals can be observed in Fig. 8. Consequently,

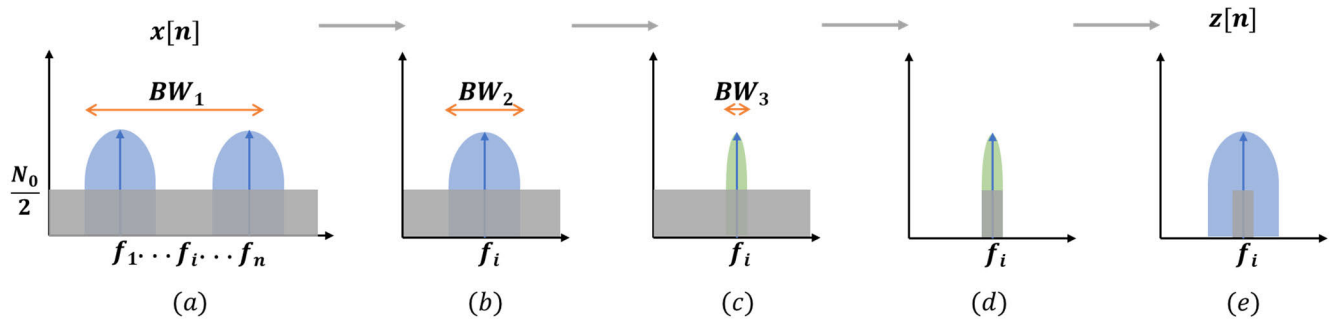


FIGURE 9. Progression of noise suppression in dynamic filtering process from a systems perspective. (a) Signal spectrum with multiple center frequencies. (b) Narrowing in a single frequency. (c) Signal phase is nullified to decrease the bandwidth. (d) Excess noise is filtered out. (e) Phase is recorrected, thus resulting in less noise captured. Adapted from [25].

the narrower the desired bandwidth, the longer it takes for the system to settle to a steady state. For standard communications systems in which the bandwidth occupies less than 5 percent of the carrier frequency, this settling time may cost a significant portion of the initial symbols after the transition, which makes traditional filter switching inadequate for real time symbol-level switching and filtering. Therefore, the preservation of charge in dynamic switching introduces a seamless way to switch between passbands, which would be especially desirable in frequency-modulated signals.

The beginning portion of the phase-sensitive dynamic filter shown in Fig. 7 consists of a multiplier and an integrator. It can thus also be considered as a correlator between the incoming signal and the desired phase and frequency modulation sequence, with the addition of the phase correction at the output to recover the waveform that falls into the correlation.

D. NOISE IMPACT

In a typical communications system, noise should always be considered. Traditionally, for a fixed-bandwidth approach to designing a bandpass filter such as that depicted in Fig. 1(a), the captured noise power that remains unfiltered out is as follows:

$$P_{N1} = \frac{B_{N1} \cdot N_0}{2} = ((f_1 - f_2) + \Delta f_{eff}) \cdot \frac{N_0}{2} \quad (18)$$

where P_{N1} is the captured noise power, f_1 and f_2 are the center frequencies of the signal, N_0 is the noise spectral density, B_{N1} is the bandwidth of the filter, and Δf_{eff} is the effective bandwidth corresponding to the effective symbol rate (which is dependent on the shape of the pulse imposed on the signal). For traditional switched filters with passbands that follow the respective frequency of propagation, the captured noise power is as follows:

$$P_{N2} = \Delta f_{eff} \cdot \frac{N_0}{2}. \quad (19)$$

As can be observed, the captured noise power is lower and does not include the frequency range across the two carriers between which the signal hops. Finally, for the dynamic filtering approach, assuming knowledge of the frequency or

phase modulated signal a priori, the amount of noise power follows the expression:

$$P_{N3} = \Delta f_{dyn} \cdot \frac{N_0}{2} = \frac{\ln |b'|}{-\pi} \cdot \frac{N_0}{2} \quad (20)$$

where Δf_{dyn} is the bandwidth of the dynamic passband, which is inverse of the time constant that can be derived from (5), and b' is the complex feedback coefficient in the integration process given in (14). As mentioned before, the narrower the bandwidth, the longer the time constant of the filter – which is desired. Additionally, after the phase correction, the bandwidth of the signal becomes compressed, and the passband only needs to be selected such that it sufficiently captures the sinusoids of the varying frequencies. Thus, the bandwidth corresponding to how much noise is carried through the filtering process can ideally be limited directly down to a fraction of the value, only passing through noise which falls into the correlation band of the desired signal. After correcting the phase, the resulting noise is at least less than that of the traditional switched bandpass filters. This noise-filtering process can be visualized in Fig. 9.

E. PHASE-SENSITIVE DYNAMIC NOTCH FILTER MODEL

In the same way that lowpass filters can be transformed to bandpass filters using complex coefficients, highpass filters can also be transformed to form notch (or bandstop) filters in the same manner. Similarly, only a single-pole complex IIR highpass filter will be covered. The typical single-pole digital highpass IIR filter takes the following form

$$y[n] = a_0 x[n] + a_1 x[n-1] + b_1 y[n-1] \quad (21)$$

$$a_0 = \frac{1+d}{2}, a_1 = -\frac{1+d}{2}, b_1 = d. \quad (22)$$

As can be observed from (21), there are two feedforward terms in this model, in contrast to the single term for dynamic bandpass filtering. For single-pole IIR filters, the values of a and b are typically chosen based on the desired normalized passband frequency, calculated around the same decay value, d . Using the z-transform of the one-pole filter, the transfer

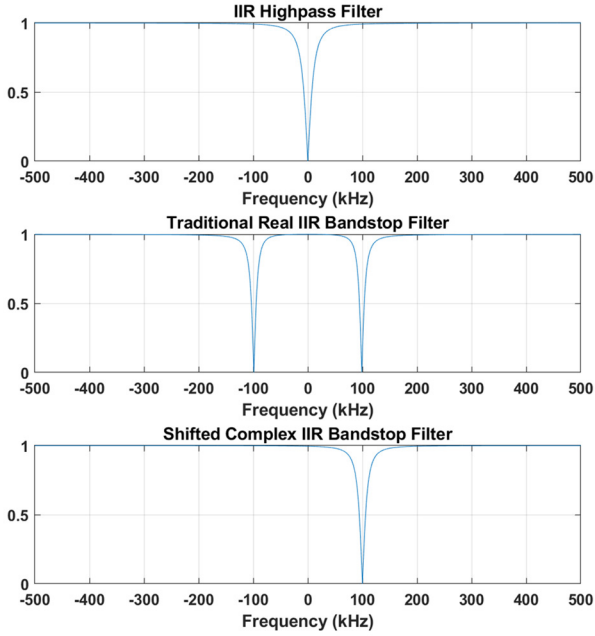


FIGURE 10. Comparison of frequency responses across lowpass filters, traditional bandpass filters, and complex IIR filters.

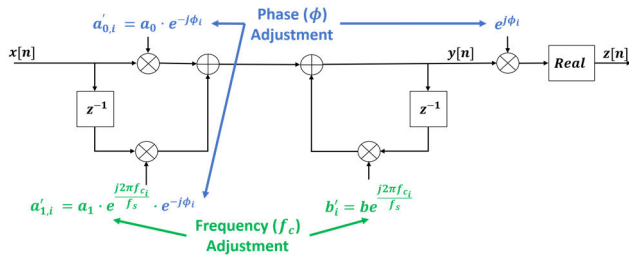


FIGURE 11. Full phase-sensitive dynamic notch filtering functional block diagram.

function can be solved algebraically as:

$$Y[z] = a_0 X[z] + a_1 X[z] z^{-1} + b_1 Y[z] z^{-1} \quad (23)$$

$$H[z] = \frac{Y[z]}{X[z]} = \frac{a_0 + a_1 z^{-1}}{1 - b_1 z^{-1}}. \quad (24)$$

In the same manner as before, by substituting in $z_{shift} = e^{-j\omega_c T} z$ into (24), we can effectively shift the center stopband frequency from 0 to ω_c . This produces the following equation

$$H_{complex}[z] = \frac{a_0 + a_1 e^{j\omega_c T} z^{-1}}{1 - b_1 e^{j\omega_c T} z^{-1}} = \frac{a_0 + a'_1 z^{-1}}{1 - b'_1 z^{-1}} \quad (25)$$

$$\begin{aligned} b'_1 &= b_1 e^{j\omega_c T} = b_1 e^{j2\pi \left(\frac{f_c}{f_s}\right)} \\ a'_1 &= a_1 e^{j\omega_c T} = a_1 e^{j2\pi \left(\frac{f_c}{f_s}\right)}. \end{aligned} \quad (26)$$

where b'_1 and a'_1 are complex coefficients, and $T = 1/f_s$ is the sampling interval. Evidently from (26), we see that the complex frequency shift exists in both the feedforward term and the feedback term. The asymmetry about the zero-frequency axis is also an indication of this filter's complex nature.

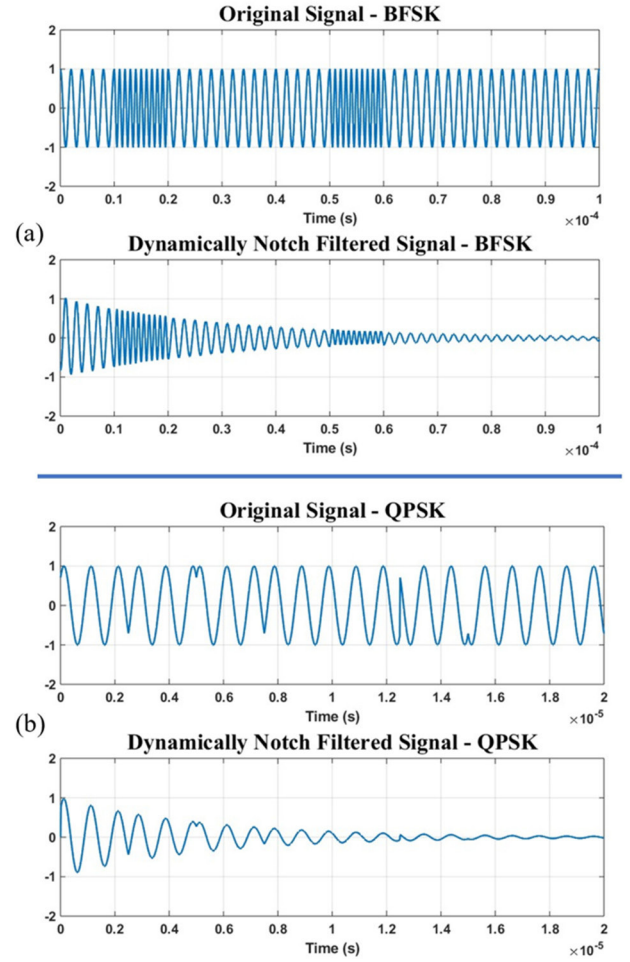


FIGURE 12. (a) Continuous charging effect of modulated BFSK signals and (b) QPSK signals in dynamic filtering.

In the same way as before, the symmetry and bandwidth of the entire frequency response about the center frequency is preserved. This can be seen in Fig. 10. Similarly, phase-sensitivity is also present in the dynamic notch filter. As seen prior, compression is introduced in the feedforward term in order to nullify the phase, thus reducing the bandwidth of the signal while preserving the bandwidth of the passband. Reconstruction of the phase is introduced in the final portion of this filtering process. The same exponential multiplier is added. This can be visualized in the time domain:

$$y[n] = a'_0 x[n] + a'_1 x[n-1] + b'_1 y[n-1] \quad (27)$$

$$\begin{aligned} a'_0 &= e^{j\phi_i} a_0 \\ a'_1 &= e^{j\phi_i} e^{j\omega_c T} a_1 \\ b'_1 &= e^{j\omega_c T} b_1 \end{aligned} \quad (28)$$

$$z[n] = \text{Re} \left\{ e^{j\phi_i} y[n] \right\}. \quad (29)$$

The full process is illustrated in Fig. 11. However, the difference here is that since we are only filtering the positive

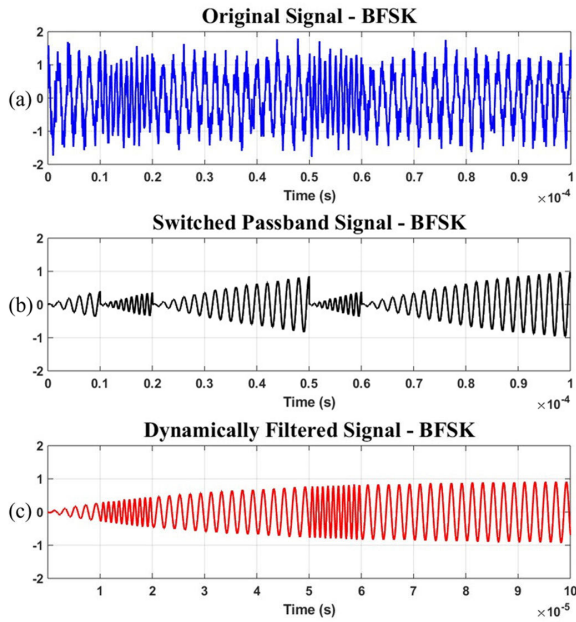


FIGURE 13. Time-domain comparison between switched-passband methods, and dynamically filtered methods utilizing the same bandwidth across BFSK signals.

frequencies of a real signal, the negative frequencies corresponding to that same signal will be observed in the output. Because of this, simply taking the real component would not present us with the desired result. The goal here is to demonstrate dynamic notch filtering in a way that highlights the “continuous decaying” nature of dynamic notch filters, paralleling the way “continuous charging” nature of dynamic bandpass filters. Thus, complex single-sideband signals are sent through this filtering process instead. This way, in taking the real output, we can observe the desired filtering effect in the time-domain in Fig. 12.

III. SIMULATIONS WITH NOISY BFSK AND QPSK SIGNALS

To demonstrate the performance of the phase-sensitive dynamic filters, the ideal digital filters illustrated in Fig. 7 and Fig. 11 are implemented in Simulink and MATLAB. In order to highlight the correlative and charge-preserving nature of PSDFs, noisy signals are generated and comparatively filtered. Binary frequency-shift keying (BFSK) and quadrature phase-shift keying (QPSK) are selected as representative test cases to isolate frequency tunability and phase sensitivity, respectively, under constant-envelope conditions. These modulation formats provide controlled demonstrations of the underlying dynamic behavior; however, the framework is not limited to these schemes and can be extended to higher-order phase and frequency modulation schemes.

A. BFSK SIGNALS

BFSK signals are generated with two carrier frequencies of $f_{c,1} = 1\text{MHz}$ and $f_{c,2} = 0.5\text{MHz}$, corresponding to each

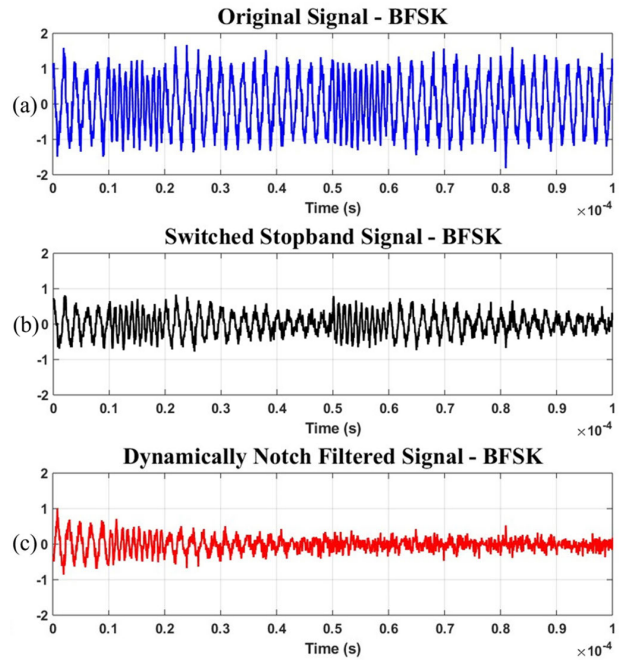


FIGURE 14. Time-domain comparison between switched-stopband methods, and dynamically notch filtered methods utilizing the same bandwidth across BFSK signals.

symbol. The SNR of the generated signal is 10 dB. The signal bandwidth is given to be $BW = 0.1\text{MHz}$ with a sampling rate of $f_s = 16\text{Msps}$. This corresponds to a symbol period of $T_s = 10\mu\text{s}$. The bandwidth of the phase-sensitive dynamic filter is set to be 1/10 of the signal bandwidth, $\Delta f = 10\text{kHz}$. This corresponds to a time constant of $\tau_f = 31.2\mu\text{s}$, which is $\sim 3\times$ longer than the symbol period. The signal duration is 100 symbols, and the first 10 symbols are plotted in Fig. 13(a) and Fig. 14(a).

For a BFSK signal shown in Fig. 13(a), the noise can be sufficiently filtered out with a switched filter bank configuration where the signal is fed through a switch that chooses between two narrow and fixed bandpass filters centered at the two choices of frequency respectively. However, Fig. 14(b) shows that, even when the passband frequency is set to match that of the input symbol, the output values in the beginning of each frequency symbols appear to “recharge” from zero towards a certain limit. When the symbol period is shorter than the time constant of the narrow bandpass filter, this charging process cannot be completed, which will exhibit distortions and energy loss in the output waveforms.

To combat this, we apply the dynamic filtering method. As mentioned before, each pair of coefficients are individually utilized at instances in time in which their corresponding carrier frequencies are present. For example, when symbol ‘1’ is transmitted and f_1 is the carrier frequency, we apply the coefficients with a passband frequency corresponding to f_1 . This contrast is evident in Fig. 13. For dynamic filtering, the “charge” for the filtered signals is preserved between

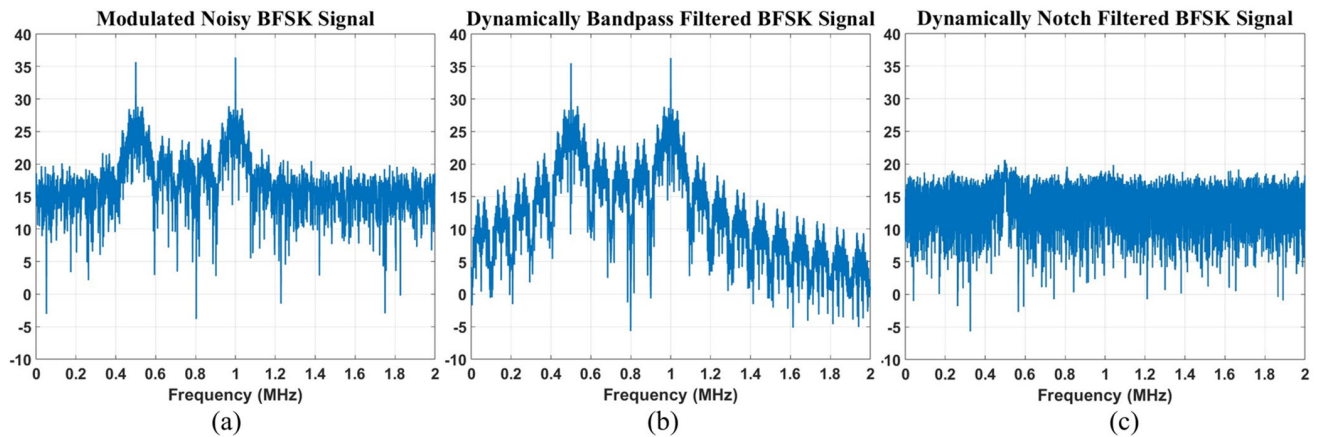


FIGURE 15. Frequency spectrum of the (a) noisy BFSK signal, (b) after dynamic filtering, and (c) after dynamic notch filtering. The signal frequencies are either passed through or attenuated.

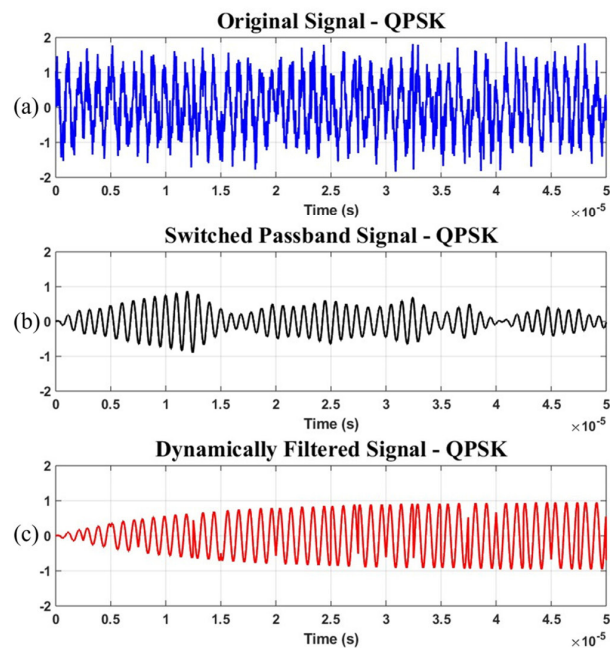


FIGURE 16. Time-domain comparison between switched-passband methods, and dynamically filtered methods utilizing the same bandwidth across QPSK signals.

frequency shifts as shown in Fig. 13(c). Essentially, there is no discharging effect between frequency switches, as explained in the previous section. This contrasts with the switched bandpass filter, in which the filter must “reset” with the switching of each symbol or passband due to a lack of charge-preservation. The spectrum of such a signal before and after the dynamic bandpass filtering process can be seen in Fig. 15(a,b). As showed in the figure, the noise at frequencies between the two carrier tones is attenuated without warping the shape of the signal tone – a trait not seen in traditional switched bandpass filter procedures.

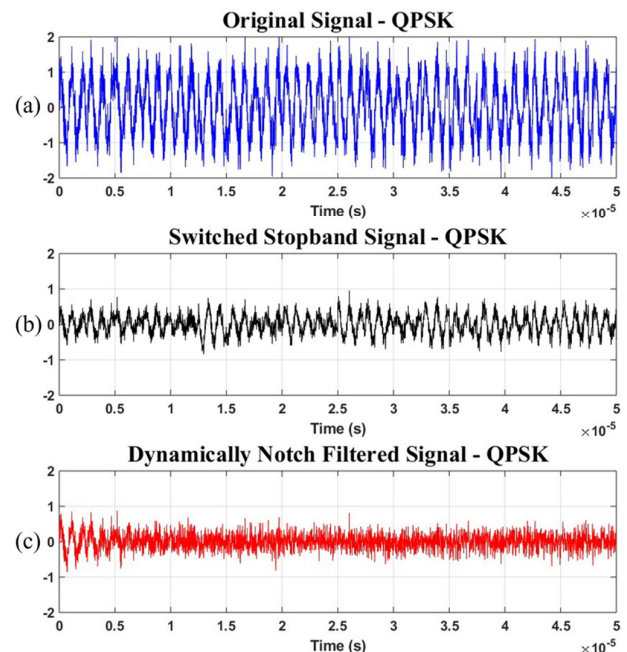


FIGURE 17. Time-domain comparison between switched-stopband methods, and dynamically notch filtered methods utilizing the same bandwidth across QPSK signals.

Similarly, dynamic notch filtering can also be performed using the same dynamic filtering procedure. As can be observed in Fig. 14, simply switching the coefficients at their respective instances in time results in the “continuous decaying” nature, whereas the switched stopband filter must restart its decay with each frequency shift. The spectrum of such a signal after the dynamic notch filtering process can be seen in Fig. 15(c). The dynamic notch filtering process only filters out the tones correlated to the stopband, which is dictated by the coefficients in the time domain. It is noted

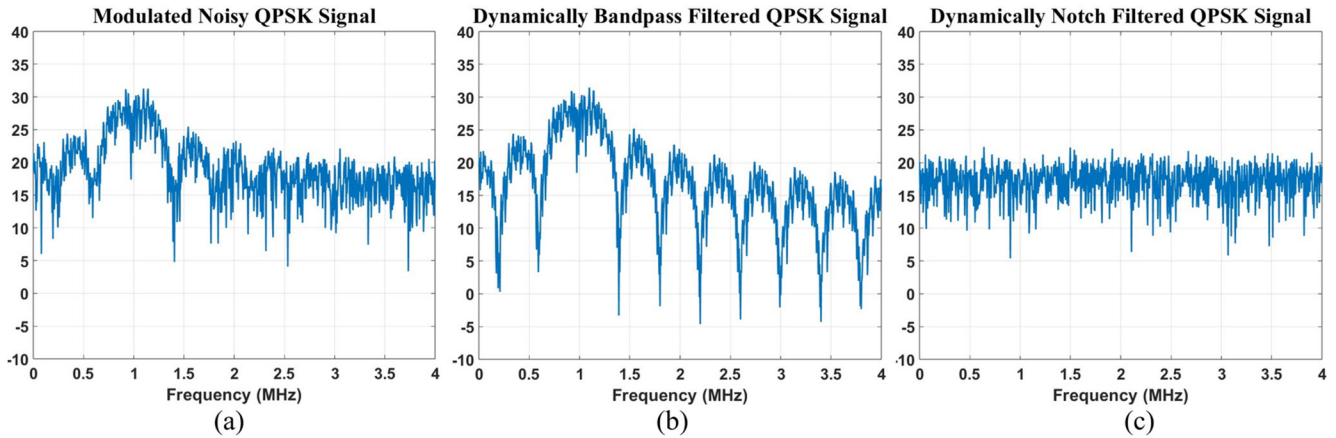


FIGURE 18. Frequency Spectrum of the (a) noisy QPSK signal, (b) after dynamic filtering, and (c) after dynamic notch filtering. The signal frequencies are either passed through or attenuated.

that no noise is suppressed by the dynamic notch filter, as it is uncorrelated in nature even at the same frequency band.

B. QPSK SIGNALS

This same comparison is performed on noisy QPSK-generated signals as shown in Fig. 16(a). QPSK signals are generated at a carrier frequency of $f_c = 1\text{MHz}$. The SNR of the generated signal is 10dB. The signal bandwidth is given to be $BW = 0.4\text{MHz}$ with a sampling rate of $f_s = 32\text{Msps}$. This corresponds to a symbol period of $T_s = 2.5\mu\text{s}$. The bandwidth of the dynamic filter is set to be $\Delta f = 30\text{kHz}$. This corresponds to a time constant of $\tau_f = 10.4\mu\text{s}$, which is $\sim 4x$ longer than the symbol period. The signal duration is 200 symbols, and the first 20 symbols are plotted in Fig. 16(a) and Fig. 17(a).

For these signals, the phase is no longer continuous across the entire duration. Using a fixed and narrow bandpass filter, the noise can be sufficiently filtered out yet the signal is distorted and with energy and information lost when the bandwidth of the filter is set to narrower than the bandwidth of the QPSK modulation, as evidenced by Fig. 16(b). Similar to the behavior previously seen, the initial symbol will “charge” towards a certain limit since the passband and the instantaneous signal frequency match. However, at the symbol transitions, the phase discontinuity results in a mix of incomplete discharge of the previous symbol and recharge of the current symbol, which drops the magnitude and warps the phase of the output.

When dynamic filtering is performed, rather than modifying the feedback term, the phase of the feedforward coefficient, a' , is switched corresponding to the phase shifts in the transmitted QPSK signal. The precise phase changes and signal envelope lost in the filtering process of a fixed bandpass filter are preserved with the dynamic filtering method in Fig. 16(c). Once more, the spectrum of the signal before and after the dynamic bandpass filtering procedure can be seen in Fig. 18(a,b).

Likewise, dynamic notch filtering can also be performed to effectively filter out the signal with correlating symbol transitions. This can be seen in Fig. 17. The fixed notch filter cannot remove the signal completely as shown in Fig. 17(b) while the continuous decay is observed once more, leaving no trace of the signal and only the uncorrelated noise in Fig. 17(c). The spectrum of such a signal after the dynamic notch filtering process can be seen in Fig. 18(c), showing a similar behavior to the notch-filtering behavior applied on the BFSK signal.

C. AMPLITUDE-MODULATED INTERFERERS

To evaluate the robustness of phase-sensitive dynamic filtering to blockers with varying envelopes, a simulation is performed in which a desired QPSK waveform is corrupted by an in-band amplitude-modulated (AM) interferer at the same carrier frequency. The desired QPSK signal uses a carrier frequency of $f_c = 1\text{MHz}$, and a modulation bandwidth of approximately $BW = 0.4\text{MHz}$, with a symbol period of $T_s = 2.5\mu\text{s}$ and a sampling rate of $f_s = 64\text{Msps}$. The bandwidth of the dynamic filter is set to be $\Delta f = 10\text{kHz}$. The phase-sensitive dynamic bandpass filter is configured with a fixed narrow bandwidth of $\Delta f = 10\text{kHz}$. This corresponds to a time constant of $\tau_f = 31.2\mu\text{s}$, which is $\sim 13x$ longer than the symbol period.

An AM interferer is then generated at the same carrier frequency and modulation bandwidth with a power level 5 dB below the desired QPSK signal. Figures 19(a,b) show the composite received waveform and the AM interferer, respectively, and Fig. 19(d) plots their spectra. After dynamic bandpass filtering, the recovered output waveform is shown in Fig. 19(c), with its output spectrum in Fig. 19(e). The results demonstrate that, when Δf is sufficiently small (equivalently, when τ_f is sufficiently long relative to the symbol period), the dynamic filtering process suppresses the AM blocker despite its time-varying envelope. Consistent with (20), narrowing the fixed bandpass bandwidth

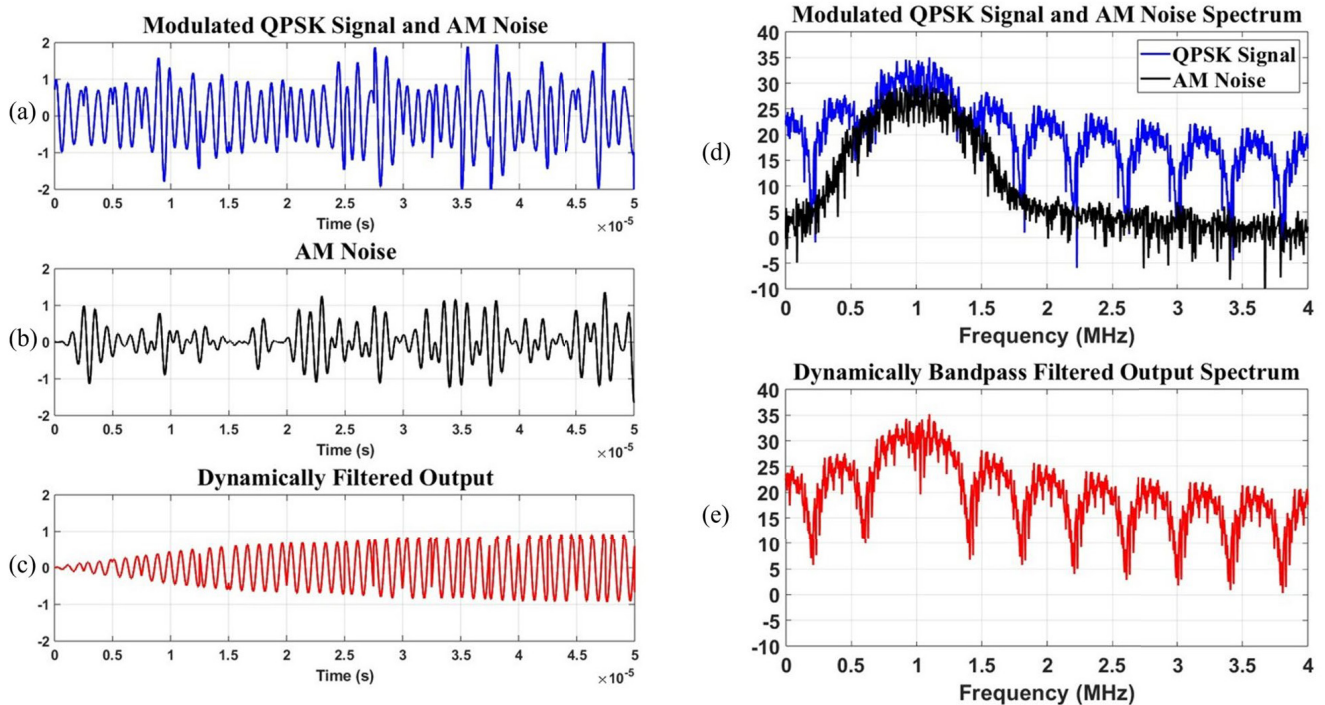


FIGURE 19. Suppression of an in-band amplitude-modulated interferer using phase-sensitive dynamic filtering. (a) Received QPSK signal with AM interference, (b) AM interferer alone, (c) dynamically filtered output, (d) spectra before filtering, and (e) spectrum after filtering.

reduces the effective in-band noise and interference power passed to the output, illustrating the tradeoff between selectivity and dynamic response governed by the filter time constant.

IV. APPLICATIONS IN CO-SITE INTERFERENCE SUPPRESSION FOR FULL-DUPLEX RADIOS

In order to maximize the full potential of phase-sensitive dynamic filters for excess noise suppression and narrow-bandwidth filtering, both the phase and frequency of the signal must be known at every moment in time. Essentially, there must be a priori knowledge on what the transmitted signal is before filtering it out, which is not always the case. One application leveraging the potential of PSDFs is pertaining to the suppression of co-site interference of communications systems, in which information on the transmitted signal is indeed presently available. Such a filter could even be implemented in the front-end of full-duplex transceiver systems to suppress the transceiver leakage. Full-duplex radio systems are capable of simultaneously transmitting and receiving at the same time and frequency [37], [38], [39], [40]. Unfortunately, the drawbacks of such systems usually take the form of leakage from the transmit-path into the receive-path. This is especially detrimental for the common case of occupancy within the same channel.

Both digital and analog implementations of leakage-cancellation structures exist for full-duplex systems to realize high isolation levels [40], [41], [42], [43]. Traditionally, even within analog implementations, there are a diverse

range of approaches that can be taken to achieve high isolation. Utilization of antenna diversity, antenna array placement and circulators are a few examples. An open-loop or closed-loop leakage cancellation scheme is often needed for additional isolations. In leakage cancellation schemes, one feeds through a part of the transmitted RF waveform and adjusts its magnitude and phase such that it cancels the transceiver leakage. From one of the first published works in adaptive leakage cancellation, it is found that the precise match of phase between the feedthrough path and the leakage path often requires an error of less than 3° at the RF carrier frequency for good cancellation performance [43]. As the phase of the leakage path is altered by the antenna and its surroundings, adjusting the phase adaptively is often required yet achieving good cancellation performance over a broadband is still challenging.

With the phase-sensitive dynamic filter, the effectiveness of such an isolation technique increases as there is no need for precise phase matching of RF signals between the two paths. The mechanism behind this is to (1) apply a dynamic bandpass filter to the received signal that contains the leakage to extract only the leakage signal without the desired signal and (2) to cancel the leakage from the received signal. Since both the extracted leakage and the received signals are from the same path, there will be no concern of delay/phase mismatch between the two. Alternatively, one can also apply a dynamic notch filter that takes the leakage signal out from the received signal with the a priori knowledge of the leakage signal frequency and phase.

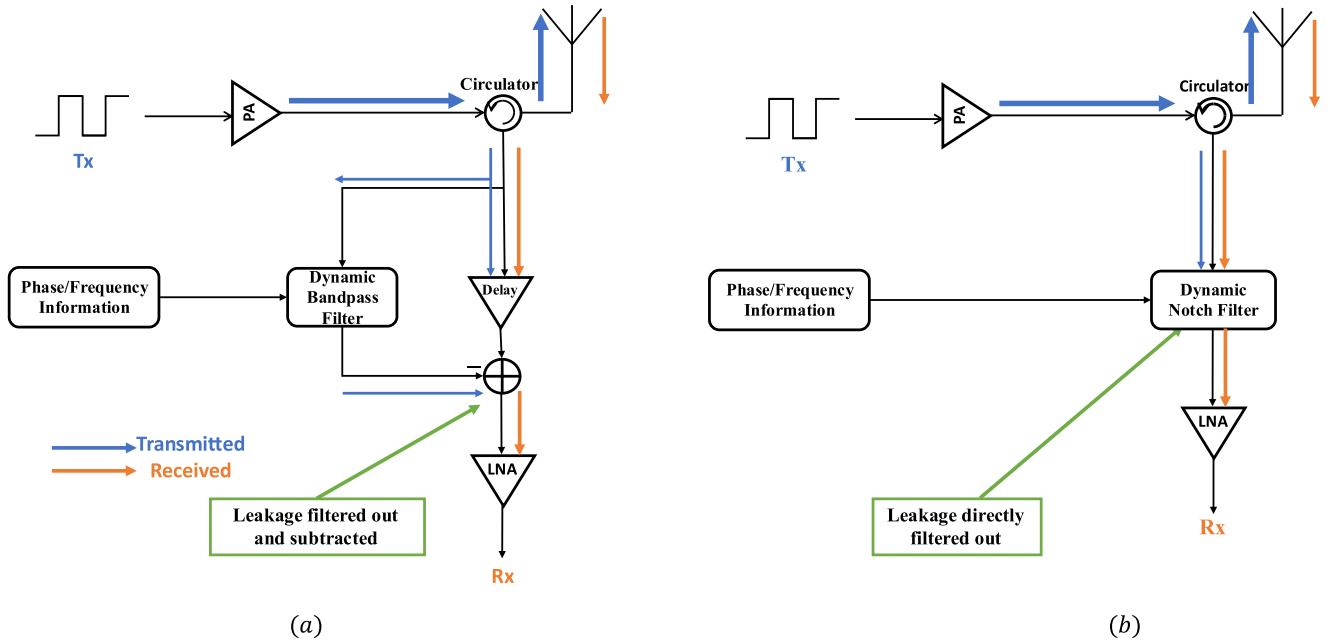


FIGURE 20. Schematic of a typical front-end STAR system with imperfect isolation between the transmit and receive path. Presented are two configurations utilizing the dynamic bandpass filter (a) and dynamic notch filter (b) to suppress the leakage from the transmitter path.

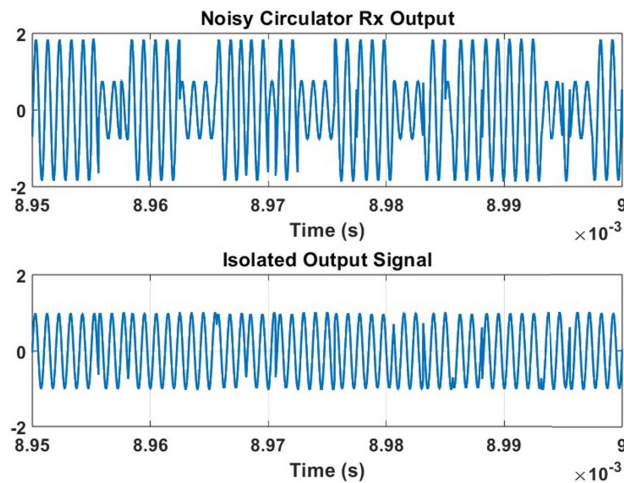


FIGURE 21. Combination of uncorrelated received and leaked QPSK signals at the circulator output (top), and the isolated received QPSK signal after dynamic filtering has taken place (bottom).

Both configurations are presented utilizing the bandpass and notch filters in Fig. 20. For the dynamic bandpass filter, the changing frequency/phase stamps are gathered from the transmitted signal to filter the received signal with those same coefficients. That filtered signal can then be used to isolate the leaked portion of the transmitted signal which incidentally makes its way through the circulator. After isolation, the phase of that isolated error signal can be flipped and then subtracted from the overall received signal before reaching the LNA. Ideally (without considering added noise), this should be able to cancel out any leakage from the transmitted signal across to the received signal. For the notch filter,

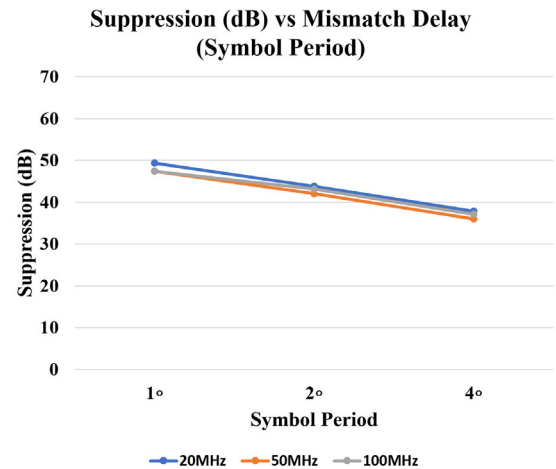


FIGURE 22. Suppression (dB) vs. Mismatch Delay (with respect to symbol period) across 20MHz, 50MHz, and 100MHz baud rate and pulse-shaped symbols. The signals undergo are pulse shaped. Reproduced from [25].

the configuration is much simpler. The dynamic notch filter gathers the changing frequency/phase stamps in the same way as before, but this time to directly filter out any leakage correlating to those changing coefficients. The key here is the correlative nature behind the phase-sensitive dynamic filters – only signals correlating to the exact frequency and phase will play a role in upkeeping the existing charge or decay.

This method of filtering bypasses the need for any adaptive amplitude or phase adjustments, as the leaked portion correlating to what is transmitted is directly filtered out from the received signal. We observe that leakage even within the same frequency band as the received signal can be filtered

out. A simple simulation is performed in Simulink using a QPSK modulated signal for the received signal and an uncorrelated QPSK modulated signal for the transmitted signal at the same frequency and amplitude. The simulation results shown in Fig. 21 indicate that in an ideal scenario (i.e. with a sufficiently long filter time-constant), all the interference can be filtered out even for two signals operating at the same frequency due to the phase-coherence nature of dynamic filtering.

With this filtering procedure, the most important variable is the coefficient mismatch delay. That delay is mimicked in this model by intentionally mistiming the changing of coefficients relative to when the symbol transition actually occurs in the input signal. This is because the changing of coefficients would have to be delayed according to the delay of the leakage signal to properly suppress the leakage. It is also the greatest source of concern, as the concept of dynamic filtering assumes perfect passband/phase adjustments in accordance with the exact instances in time when they occur for the filtered signal. As with all physical systems, imperfections will exist, and the coefficient mismatch delay should be accounted for. It will be shown, however, the tolerance of the delay mismatch with dynamic notch filter is much more relaxed comparing to that in the standard leakage cancellation scheme as it is determined by the symbol period instead of the period of RF carrier.

Theoretically, we can obtain the results for suppression under an artificially induced mismatch in Fig. 22. As can be observed in that plot, the theoretical suppression still remains far above 30 dB (typical leakage value of a state-of-the-art circulator) at a 4° mismatch with respect to symbol period. Visually, symbol-transition regions are the regions which are most adversely affected by coefficient mismatches; but for pulse-shaped signals, this region is attenuated due to the symbol-shaping effect. Practically speaking, pulse-shaped signals also exist in any form of band-limited hardware in the front-end and are shown to improve the theoretical suppression limit in the presence of coefficient mismatch delay.

V. DISCUSSION ON LIMITATIONS OF PHYSICAL IMPLEMENTATIONS

The phase-sensitive dynamic filtering (PSDF) framework presented in this work is formulated at an ideal, behavioral level. Practical circuit realizations will inevitably introduce non-idealities that affect performance.

A. PHYSICAL REALIZATIONS

The closest physical realization to PSDF is the parametric filter based on cascaded parametric mixers [17], previously demonstrated for continuously tunable bandpass filtering. While the architecture was not designed for symbol-level passband switching, it demonstrates that narrowband, energy-preserving filtering devices utilizing time-varying mixing operations are physically realizable. Extending such structures towards PSDF operation would require tighter

synchronization between the control (pump) signals and incoming waveform.

B. PHYSICAL LIMITATIONS AND CONSIDERATIONS

Nonlinearity is a key concern for physical realizations of PSDFs, particularly in the presence of strong out-of-band interferers. Large blockers may intermodulate through nonlinear mixing processes, generating in-band distortion products that are not captured by the idealized model presented here. As a result, practical PSDF implementations would need to exhibit high linearity and robust power handling, especially at the front end, to preserve interference rejection performance.

Timing jitter and phase noise in the control path further limit practical realizations, particularly at higher carrier frequencies. Constant symbol-delay mismatch was analyzed in Fig. 22, where modest timing offsets were shown to degrade suppression gradually rather than abruptly. In addition to such deterministic offsets, timing jitter and phase noise introduce mismatch that reduces phase coherence and weaken the energy-accumulation mechanism central to PSDF operation. In the presence of strong blockers, this effect is analogous to reciprocal mixing in conventional receivers, raising the in-band noise floor. Importantly, this limitation is not unique to PSDFs but is inherent to any architecture employing time-varying mixing. These considerations highlight the need for low-jitter control sources and carefully designed synchronization architectures in future implementations.

Switching transients and spurious content are also well-known challenges in N-path and switched-filter-bank architectures. Such effects would directly deteriorate the theoretical noise suppression limits predicted by the PSDF model, introducing additional noise into the signal path. However, parametric-mixing-based implementations offer a distinct advantage in this regard and represent an optimistic path forward. Rather than relying on hard switching, parametric devices utilize continuous modulation of circuit parameters, such as capacitance or inductance. In these systems, the dominant spurious content arises from leakage of pump and idler frequencies, rather than broadband switching noise.

The impact of these non-idealities is strongly topology-dependent. The extent to which energy preservation and theoretical noise suppression limits are degraded will vary with the specific implementation architecture, device technology, and synchronization scheme. Accordingly, quantitative deviation from the ideal PSDF model must be evaluated on a per-implementation basis.

VI. CONCLUSION

This work presents an idealized theoretical framework that uses time-varying complex-coefficient IIR filters as a modeling abstraction to formalize the behavior of phase-sensitive dynamic filters (PSDFs). Rather than introducing a new realizable filter topology, the framework employs complex IIR filters to capture the essential operating principles of

symbol-synchronous dynamic filtering with long time constants. Through theoretical analysis and simulation, the model illustrates how modulation of the coefficient phase enables rapid control of a filter's passband and phase response while maintaining amplitude continuity of the waveform, resulting in coherent filtering across multiple symbols without collapse of the output envelope.

While the analysis in this paper is carried out at the system level, charge-preserving devices with phase-sensitive and rapidly tunable behavior similar to that modeled here have already been demonstrated in existing parametric and time-varying RF technologies. However, a unified theoretical model capturing these behaviors in a form suitable for system-level analysis has been lacking. The framework presented in this work provides such a model, offering a foundation that formalizes and extends prior physical observations.

From a system perspective, the results suggest that PSDFs offer new opportunities for front-end signal conditioning in communication systems, including applications such as leakage suppression and agile frequency operation. By enabling symbol-synchronous switching while maintaining long time constants, the modeled behavior highlights a pathway toward improved noise selectivity and enhanced spectral utilization in dynamically changing environments. Incorporating implementation-specific non-idealities and experimental validation of the phase-sensitive dynamic filtering behavior remain important directions for future work.

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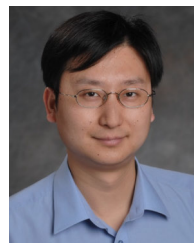


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